MYSQL 8.0+ ENTERPRISE ANALYTICS

Walmart Sales Data Analysis & SQL Engineering

End-to-end database pipeline covering schema architecture, feature engineering, complex joins, views, dynamic stored procedures, and automated database triggers.

8 Implementation Modules

Automated Triggers & Stored Procedures

Executive Overview

Project Scope

Relational Integrity

Standardized sales records with target revenue benchmarks and promotion discount mappings.

Business Intelligence

6 pre-computed analytical views delivering real-time metrics on branch ratings, basket sizes, and profitability.

Engine Automation

Active database triggers enforcing casing rules, automatic total re-calculation, and audit trail generation.

Database Schema Topology

PRIMARY ENTITIES

sales (Core Fact Table)
invoice_id, branch, city, product_line, unit_price, quantity, total, cogs, VAT, date...

branch_targets (Dimension Table)
branch (PK), city, target_revenue, manager_name

promotions (Dimension Table)
product_line (PK), discount_percentage, promo_code

Entity Relationship Design

Structured around a high-performance star-like schema to optimize aggregate queries, joins, and reporting performance without data redundancy.

Audit & Compliance

Includes `sales_audit` table to track transaction execution events with auto-timestamps.

Feature Engineering Pipeline

Transformed raw date/time fields into structured time-series attributes for micro-trend analytics.

```sql
-- Time Shift Classification
UPDATE sales
SET time_of_day = CASE
  WHEN time < '12:00:00' THEN 'MORNING'
  WHEN time BETWEEN '12:00:00' AND '18:00:00' THEN 'AFTERNOON'
  ELSE 'EVENING'
END;

-- Temporal Extraction
UPDATE sales SET day_name = DAYNAME(date);
UPDATE sales SET month_name = MONTHNAME(date);
```

time_of_day

Enables shift-wise revenue profiling across Morning, Afternoon, and Evening.

day_name

Identifies weekly transaction spikes and peak footfall periods.

month_name

Supports monthly revenue trajectories and COGS comparison.

Business Intelligence Views

Branch Performance

Aggregates total transactions, total volume sold, total revenue, and average order value (AOV) per location.

Product Profitability

Tracks total units sold, cumulative gross income, and overall satisfaction ratings across product lines.

Customer Demographics

Evaluates basket size, purchase count, and payment method choices segmented by customer types.

Parameterized Stored Procedures

Encapsulated business logic execution with IN/OUT control parameters.

```
-- Conditional Branch Tiering Procedure
CREATE PROCEDURE evaluateperformance(
  IN input_branch VARCHAR(5),
  OUT performance_status VARCHAR(50)
)
BEGIN
  DECLARE branch_total DECIMAL(12,2);
  SELECT SUM(total) INTO branch_total
  FROM sales WHERE branch = input_branch;
  IF branch_total > 50.00 THEN
    SET performance_status = 'High Volume';
  ELSE
    SET performance_status = 'Standard Volume';
  END IF;
END//
```

Multi-Parameter Filtering

Procedural execution of complex queries across custom target parameters (Month + Customer Type).

OUT Parameter Revenue Calculation

Calculates total gross revenue per product line directly into session variables for operational workflows.

Relational Joins & Complex Filtering

Combining operational fact records with dimension tables under multi-clause logic.

INNER JOIN

Target vs Revenue

Matches sales volume against branch management target metrics.

LEFT JOIN

Promo Attribution

Evaluates promotion usage and discount applicability per item line.

UNION SIMULATION

Full Outer Join

Simulates complete outer relational coverage using left/right union logic.

Automated Database Triggers

Maintaining data integrity and automated recalculations on write events.

```
-- Before Update Auto-Recalculation
CREATE TRIGGER trg_before_update_sales
BEFORE UPDATE ON sales
FOR EACH ROW
BEGIN
  SET NEW.total = NEW.unit_price * NEW.quantity;
  SET NEW.cogs = NEW.unit_price * NEW.quantity;
  SET NEW.vat = NEW.total * 0.05;
  IF NEW.branch NOT IN ('A', 'B', 'C') THEN
    SIGNAL SQLSTATE '45000'
    SET MESSAGE_TEXT = 'Invalid branch code';
  END IF;
END//
```

Real-Time Calculation

Guarantees total, COGS, and VAT accuracy upon any edit.

Enforced Branch Validation

Throws custom exception states to block unauthorized updates.

Key Business Takeaways

8

Core SQL Analytical Modules

100%

Automated Integrity & Recalculation

6

Production Analytical Views



Walmart Sales SQL Repository

Explore the full SQL source code, setup scripts, sample query execution results, and documentation in the project GitHub repository.

MySQL 8.0+ / MariaDB 10.x Compatible